

Stock Trend & Price Prediction — Notebook Summary

Analysis of ML_project_1.ipynb: what it does, what was wrong, and what was fixed.

- 1 **Dataset:** ~50,000 rows of daily OHLCV stock data across multiple symbols, plus SMA_10/SMA_50/EMA_20, RSI, MACD, Volatility, and a Trend label (Bullish/Bearish).
- 2 **Goal:** two separate models sharing one feature pipeline — a regressor for next Close price, and a classifier for next Trend direction.
- 3 **Data quality:** a small number of rows had Close ≤ 0 (not physically possible for a stock price) and were dropped; remaining NaNs came from rolling-window indicators needing a warm-up period.
- 4 **Class balance:** Bullish vs. Bearish trend labels are roughly even, so plain accuracy is a reasonable metric for the classifier.
- 5 **Leakage control:** engineered Lag1 versions of Close, Volume, Open, High, Low, and Daily Return so the model only ever sees information available before the day it's predicting.
- 6 **Calendar features:** day-of-week, day-of-month, month, and year were added to catch any seasonal effects.
- 7 **Chronological split:** train/test split is done by time order (first 80% / last 20%), not randomly — a random split would leak future prices into training.
- 8 **Critical bug found:** the original notebook plotted and scored a variable (y_pred) for the regression model that was never actually created — the model-fitting cell was missing, even though old cached output (MSE: 1213.06) remained in the file.
- 9 **Fix applied:** added the missing Linear Regression fit/predict step back in so the notebook runs end-to-end without a NameError.
- 10 **Classifier result:** Logistic Regression on Trend_Bullish reached about 95% accuracy on the held-out chronological test set.
- 11 **Baseline regressor:** Linear Regression underfit badly — predictions clustered in a narrow band and didn't track the actual spread of prices (high MSE).
- 12 **Second feature round:** added Price_Range, Open_Close_Diff, Volume_Change, and three interaction terms (RSI×Volume, MACD×Volatility, Volatility×Daily Return) — all built from lagged or same-day-available values.
- 13 **Stronger model:** swapped in a Random Forest Regressor with the expanded feature set to capture non-linear relationships the linear model missed.

- 14 **Evaluation approach:** both models were checked with actual-vs-predicted line plots, distribution overlays, scatter plots against a 45° reference line, plus MSE/R² where relevant.
- 15 **Next steps suggested:** walk-forward cross-validation instead of one fixed split (more realistic for time series), and trying gradient boosting (XGBoost/LightGBM) as a stronger benchmark than Random Forest.